

## DEFINITION AND ANALYSIS OF THE TARGET SITUATION (T2)

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### Team Members:

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- **Transaction Processing System (TPS)**

- **DESCRIPTION OF THE NEW BUSINESS PROCESS**

*"Irish Luck Loyalty and Reward Point Transaction Process" This new process covers a "Customer Loyalty Program" where customers earn points based on their spending during each visit and convert these points into reward items. The customer presents their mobile number or digital QR ID while ordering or during payment. The system calculates the current order total and adds points at a determined rate to the customer's digital wallet as an "Input." If the customer wants to use their accumulated points, the system checks the point balance and deducts the corresponding item from the bill, removing the points as an "Output."*

- **DESCRIPTION OF COMPETITIVE ADVANTAGE**

*While most other pubs and restaurants in the area only operate with a standard sales process, thanks to this proposed system customers will choose Irish Luck again to collect points. By collecting data on consumption habits personalized campaigns can be offered.*

*Manager's Note: "We always see new people in pub but there is no fixed customer. "*

- **TECHNICAL DETAILS OF TRANSACTION PROCESSING SYSTEM**

*Hardware: An additional module that works integrated with the existing Pavo POS devices or is installed on the smartphones used by the waiters.*

*Software: A cloud-based Loyalty Management Software (SaaS). A backend service that holds the customer database and can communicate with the Pionpos API.*

*Input: Customer Phone No, Order ID (from Pionpos), Amount.*

*Output: Current Point Balance, Earned/Spent Point Transaction Record.*

*Communication: Instant data transfer via the in-restaurant Wi-Fi network.*

- **DETAILS OF USING TPS IN EXECUTING THE NEW BUSINESS PROCESS**

*1. When the customer asks for the bill, the waiter presses the "Loyalty Membership" button on the Pionpos device before going to the "Payment" screen.*

2. *The customer's phone number is entered.*
3. *The system instantly displays the customer's name and points on the screen.*
4. *If points are to be used, the relevant amount is deducted from the bill .*
5. *When the payment is completed with the Pavo POS, the newly earned points are automatically processed into the system. This process is carried out as a "Loyalty Transaction" that runs parallel to the existing sales process but is independent of it.*

- **INTEGRATION OF TPS-SUPPORTED BUSINESS PROCESS INTO EXISTING SITUATION**

*The newly proposed Loyalty TPS works directly integrated with the existing Pionpos system. In the existing "Close Order/Payment" step the system queries the Loyalty Database before sending the amount to the Pavo POS. A "Membership Check and Point Processing" sub-process is added to the current flowchart between "Request Payment" and "Payment Completed."*

- **POTENTIAL ETHICAL PROBLEMS**

*Data Privacy: Recording what customers eat/drink and their visit frequency is sensitive data under GDPR (KVKK). Explicit consent must be obtained from the customer.*

- **Management Information System (MIS)**

- **DESCRIPTION OF THE NEW BUSINESS PROCESS**

*"Data-Driven Menu and Performance Analysis Process" This is a process where management decisions are made by blending data coming from both the existing sales system (Pionpos) and the new proposed Loyalty system (Loyalty-TPS). Basic activities:*

*Data Collection: Pulling daily sales, stock usage, waiter performance, and customer loyalty data.*

*Reporting: Creating reports for "Most Profitable Hours," "Items with Highest Cost/Sales Ratio," and "Regulars Preference Analysis."*

*Decision Making (Output): Making menu price revisions, shift planning, and stock purchasing decisions based on these reports.*

- **DESCRIPTION OF COMPETITIVE ADVANTAGE**

*Thanks to the analysis of busy hours and popular products, waiter shifts are planned more efficiently, reducing customer waiting times. Recipe optimization is done by identifying which materials are wasted the most in the kitchen and bar or which products are not selling.*

- **TECHNICAL DETAILS OF MANAGEMENT INFORMATION SYSTEM**

*Software: A Business Intelligence (BI) Dashboard application*

Data: Pionpos Sales Database + New Loyalty Database + Cost/Recipe Data.

Hardware: PC or tablet in the pub.

- **DETAILS OF USING MIS IN EXECUTING THE NEW BUSINESS PROCESS**

The manager opens the MIS dashboard and pulls the "Last Week's Product Profitability" report. Low sales but high cost products are marked to be removed from the menu. By looking at the Customer Visit Hours report, the most experienced waiters are assigned to the hours when regulars are busiest.

- **INTEGRATION OF MIS-SUPPORTED BUSINESS PROCESS INTO EXISTING SITUATION**

The proposed MIS is positioned as a Monitoring and Control Layer at the top of the business. The existing Pionpos and the new Loyalty TPS systems connect to this MIS only as data providers. The MIS does not interfere with the operational flow; it informs the decisions regarding how the operational flow should be.

- **POTENTIAL ETHICAL PROBLEMS**

Monitoring waiters' sales performance and speed second-by-second can create stress on employees and cause "Micromanagement" issues in terms of business ethics.

- **PROPOSAL OF A DECISION SUPPORT SYSTEM**

The pub sector is heavily affected by special occasions and seasonality. A DSS can analyze past years' sales data, the upcoming match schedule, and the weather forecast to offer suggestions to the manager.

#### APPENDIX-I:

##### TRACEABILITY OF TARGET SITUATION TO THE EXISTING SITUATION

| EXISTING SITUATION |                | TARGET SITUATION          |                 |
|--------------------|----------------|---------------------------|-----------------|
| BP Activity        | Supporting IS  | BP Activity               | Supporting IS   |
| Table Activation   | Pionpos App    | Table Activation          | Pionpos App     |
| Order Taking       | Pionpos App    | Order Taking              | Pionpos App     |
| -                  | -              | Member Identification     | Loyalty TPS     |
| -                  | -              | Points Accrual/Redemption | Loyalty TPS     |
| Order Routing      | Pionpos Server | Order Routing             | Pionpos Server  |
| Payment            | Pavo POS       | Payment                   | Pavo POS        |
| Manager Review     | Manual         | Performance Analysis      | Integrated MIS  |
| Stock Check        | Manual         | Inventory Optimization    | Forecasting DSS |
|                    |                |                           |                 |

**APPENDIX-II:**  
**DEFINITION AND ANALYSIS OF THE CURRENT (EXISTING) SITUATION (T1)**

*NOT REVISED*

**APPENDIX-III:**

*Regarding the work distribution, Sinan Ermiş was responsible for preparing the report, while Berkay Saylak handled the interviews and facilitated the exchange of ideas with the business management. Additionally, the proposed solution was a joint effort developed collaboratively by both members.*

*Those are the photos taken during meetings.*



